

Math 162

Midterm Exam 2

Justin Campbell

1. Define the *hyperbolic sine and cosine functions* $\sinh, \cosh : \mathbb{R} \rightarrow \mathbb{R}$ by

$$\sinh x := \frac{1}{2}(e^x - e^{-x}), \quad \cosh x := \frac{1}{2}(e^x + e^{-x}).$$

- (a) Show that

$$\frac{d}{dx} \sinh x = \cosh x, \quad \frac{d}{dx} \cosh x = \sinh x, \quad \text{and} \quad \cosh^2 x - \sinh^2 x = 1.$$

- (b) Prove that for all $x \geq 0$, we have

$$\int_1^{\cosh x} \sqrt{t^2 - 1} \, dt = \frac{1}{2}(\cosh x \sinh x - x).$$

(Hint: I don't recommend evaluating the integral directly. Try applying the first fundamental theorem of calculus.)

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) := \begin{cases} \frac{\sin x}{x} & \text{for } x \neq 0 \\ 1 & \text{for } x = 0. \end{cases}$$

(a) Compute $f'(0)$.

(b) Compute $f''(0)$.

3. Let $f : [a, b] \rightarrow \mathbb{R}$ be a differentiable function such that $f' : [a, b] \rightarrow \mathbb{R}$ is integrable. Prove that

$$\lim_{\lambda \rightarrow \infty} \int_a^b f(t) \sin(\lambda t) \, dt = 0.$$

(Hint: use integration by parts.)